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NIM	: 242410101004
Kelas	: PWEB A
Program Studi	: Sistem Informasi

Tugas 1

1. Instal XAMPP terbaru, pastikan PHP 8.2+ tersedia (cek: php --version di terminal)

```
C:\laragon\www
λ php --version
PHP 8.3.30 (cli) (built: Jan 13 2026 22:50:40) (ZTS Visual C++ 2019 x64)
Copyright (c) The PHP Group
Zend Engine v4.3.30, Copyright (c) Zend Technologies

C:\laragon\www
λ
```

Pada langkah ini dilakukan pengecekan versi PHP untuk memastikan sudah menggunakan PHP versi 8.2 atau lebih, karena Laravel membutuhkan versi PHP tertentu agar dapat berjalan dengan baik.

2. Instal Composer dari getcomposer.org, verifikasi dengan: composer --version

```
C:\laragon\www
λ composer --version
Composer version 2.9.4 2026-01-22 14:08:50
PHP version 8.3.30 (C:\laragon\bin\php\php-8.3.30-Win32-vs16-x64\php.exe)
Run the "diagnose" command to get more detailed diagnostics output.
```

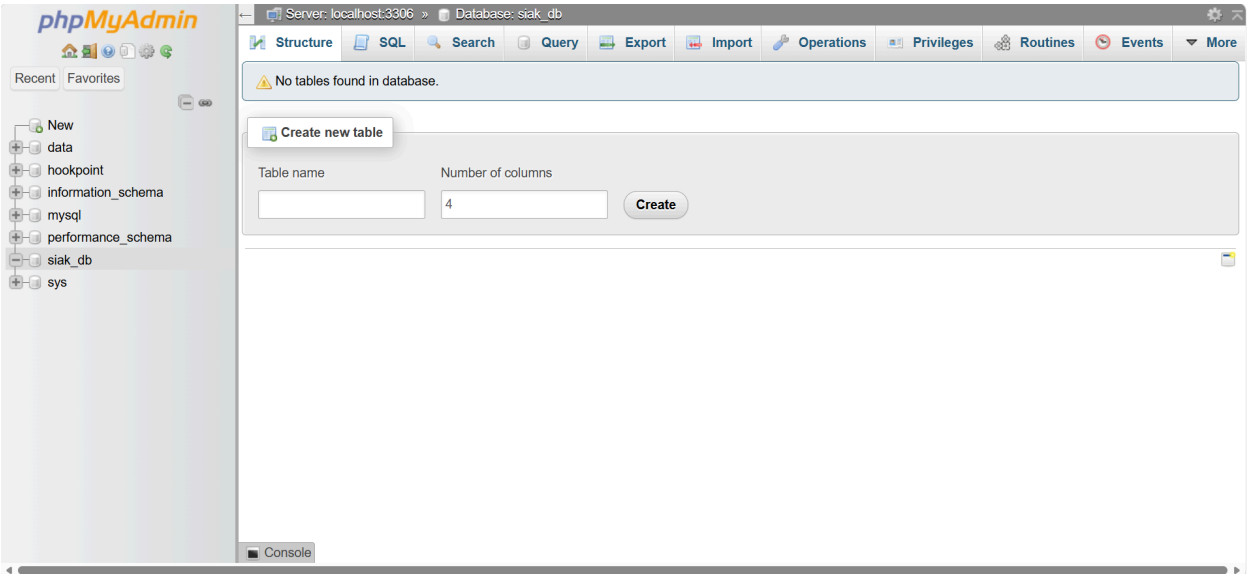
Langkah ini bertujuan untuk memastikan Composer sudah terinstall dengan benar, karena Composer digunakan untuk mengelola dependency pada Laravel.

3. Buat proyek Laravel baru: composer create-project laravel/laravel siak-app

```
C:\laragon\www
λ composer create-project laravel/laravel siak-app
Creating a "laravel/laravel" project at "./siak-app"
Installing laravel/laravel (v13.3.0)
  - Installing laravel/laravel (v13.3.0): Extracting archive
Created project in C:\laragon\www\siak-app
> @php -r "file_exists('.env') || copy('.env.example', '.env');"
Loading composer repositories with package information
Updating dependencies
Lock file operations: 107 installs, 0 updates, 0 removals
  - Locking brick/math (0.14.8)
  - Locking carbonphp/carbon-doctrine-types (3.2.0)
  - Locking dflydev/dot-access-data (v3.0.3)
  - Locking doctrine/inflector (2.1.0)
  - Locking doctrine/lexer (3.0.1)
  - Locking dragonmantank/cron-expression (v3.6.0)
  - Locking egulias/email-validator (4.0.4)
  - Locking fakerphp/faker (v1.24.1)
  - Locking filp/whoops (2.18.4)
  - Locking fruitcake/php-cors (v1.4.0)
  - Locking graham-campbell/result-type (v1.1.4)
  - Locking guzzlehttp/guzzle (7.10.0)
  - Locking guzzlehttp/promises (2.3.0)
  - Locking guzzlehttp/psr7 (2.9.0)
  - Locking guzzlehttp/uri-template (v1.0.5)
  - Locking hamcrest/hamcrest-php (v2.1.1)
  - Locking laravel/framework (v13.6.0)
  - Locking laravel/pail (v1.2.6)
  - Locking laravel/pint (v1.29.1)
  - Locking laravel/prompts (v0.3.17)
  - Locking laravel/serializable-closure (v2.0.12)
  - Locking laravel/tinker (v3.0.2)
  - Locking league/commonmark (2.8.2)
  - Locking league/config (v1.2.0)
  - Locking league/flysystem (3.33.0)
```

Pada tahap ini dibuat project Laravel baru dengan nama *siak-app* menggunakan perintah composer create-project sebagai dasar aplikasi yang akan dikembangkan.

4. Buat database baru 'siak_db' di phpMyAdmin



Database dengan nama *siak_db* dibuat di phpMyAdmin untuk menyimpan data yang nantinya akan digunakan oleh aplikasi Laravel.

5. Konfigurasi file .env: isi DB_DATABASE, DB_USERNAME, DB_PASSWORD sesuai lokal

```
DB_CONNECTION=mysql
DB_HOST=127.0.0.1
DB_PORT=3306
DB_DATABASE=siak_db
DB_USERNAME=root
DB_PASSWORD=
```

File `.env` dikonfigurasi untuk menghubungkan Laravel dengan database lokal dengan mengisi nama database, username, dan password.

6. Jalankan: php artisan key:generate

```
PS C:\laragon\www\siak-app> php artisan key:generate

  INFO  Application key set successfully.
```

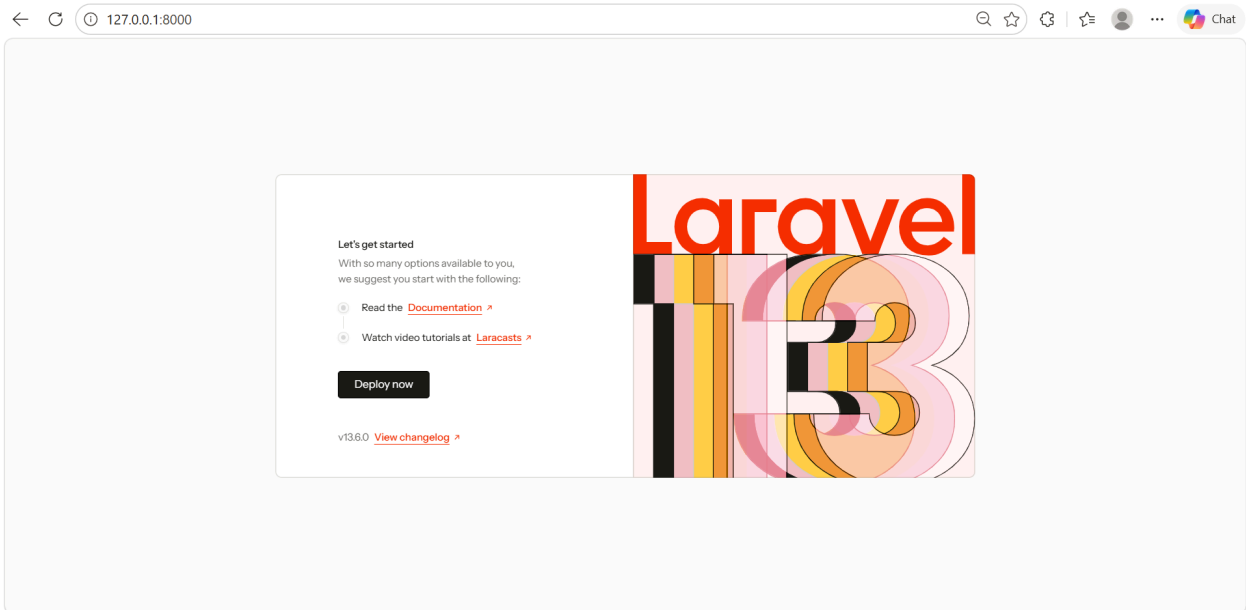
Perintah ini digunakan untuk membuat application key yang berfungsi untuk keamanan aplikasi Laravel.

7. Jalankan server: php artisan serve - buka http://127.0.0.1:8000 di browser

```
PS C:\laragon\www\siak-app> php artisan serve

  INFO  Server running on [http://127.0.0.1:8000].

  Press Ctrl+C to stop the server
```

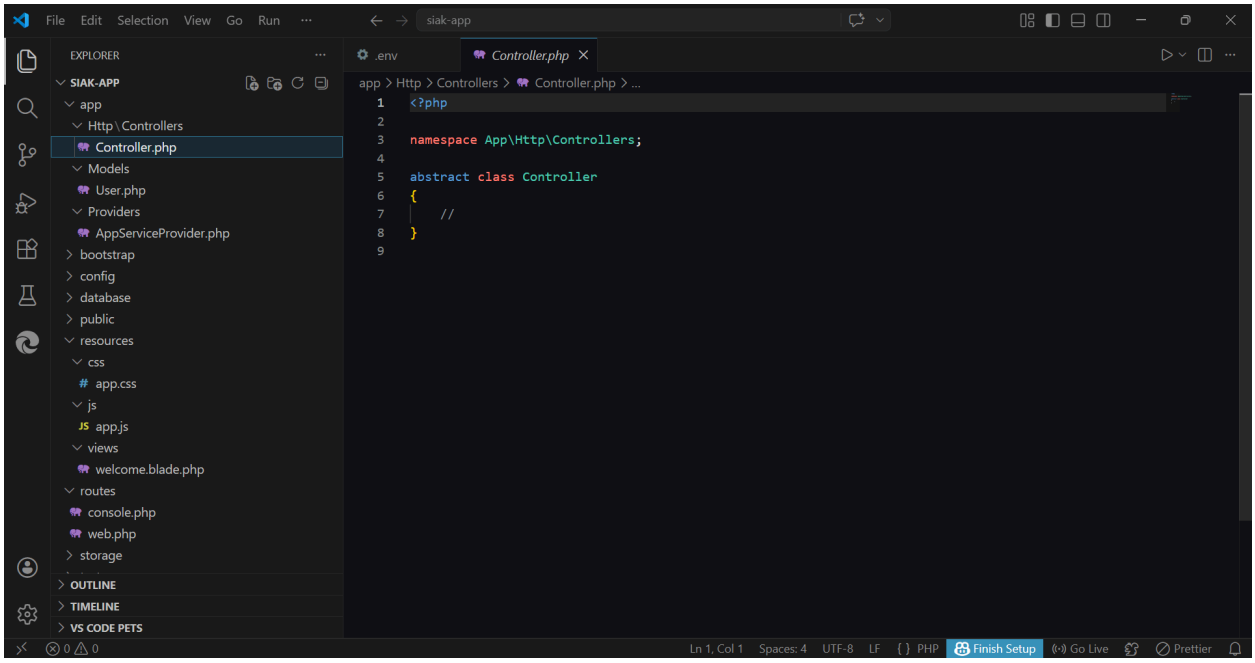


Laravel dijalankan menggunakan perintah `php artisan serve`, kemudian diakses melalui browser untuk memastikan aplikasi berhasil berjalan.

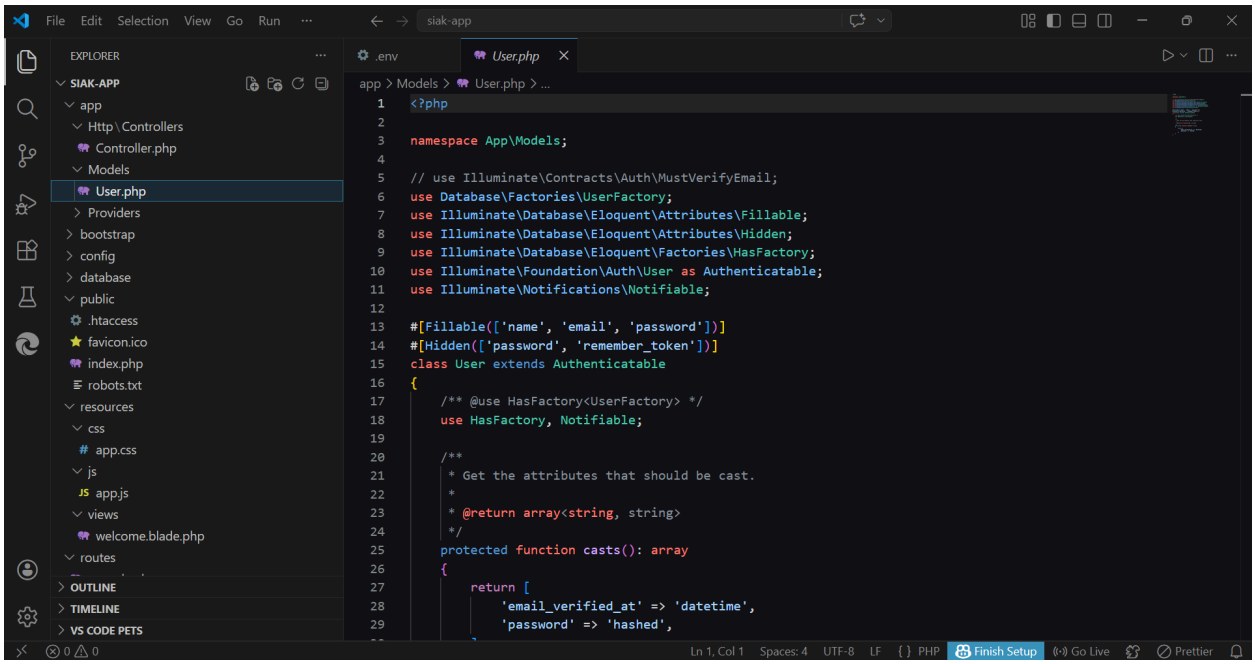
8. Eksplorasi struktur folder, buka setiap folder dan pahami fungsinya

   Pada bagian ini dilakukan eksplorasi struktur folder Laravel untuk memahami fungsi masing-masing folder. Folder seperti *app* berisi logika aplikasi, *routes* untuk pengaturan URL, *resources* untuk tampilan, dan *database* untuk pengelolaan database. Hal ini penting agar lebih mudah dalam pengembangan aplikasi ke depannya.

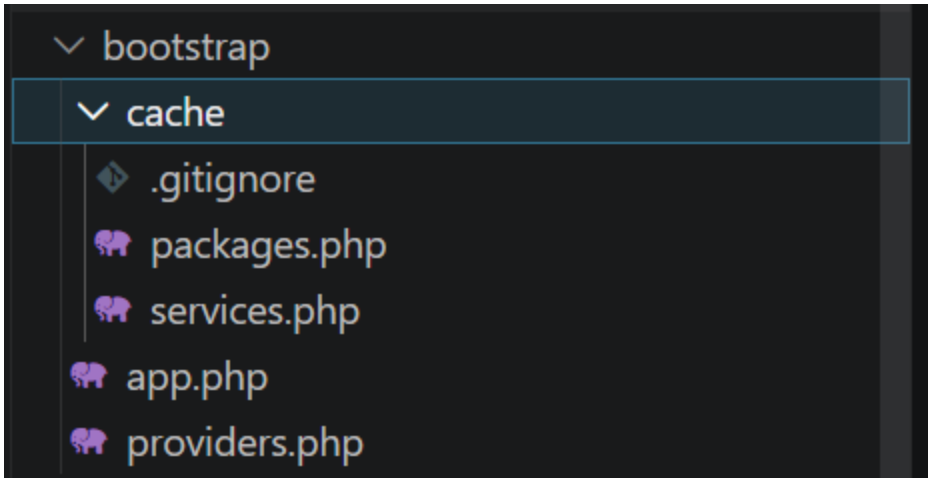
   a. app/ ← Folder ini berisi logika utama aplikasi seperti controller dan model.
   b. app/http/Controllers/Controller.php ← Controller (logika bisnis)



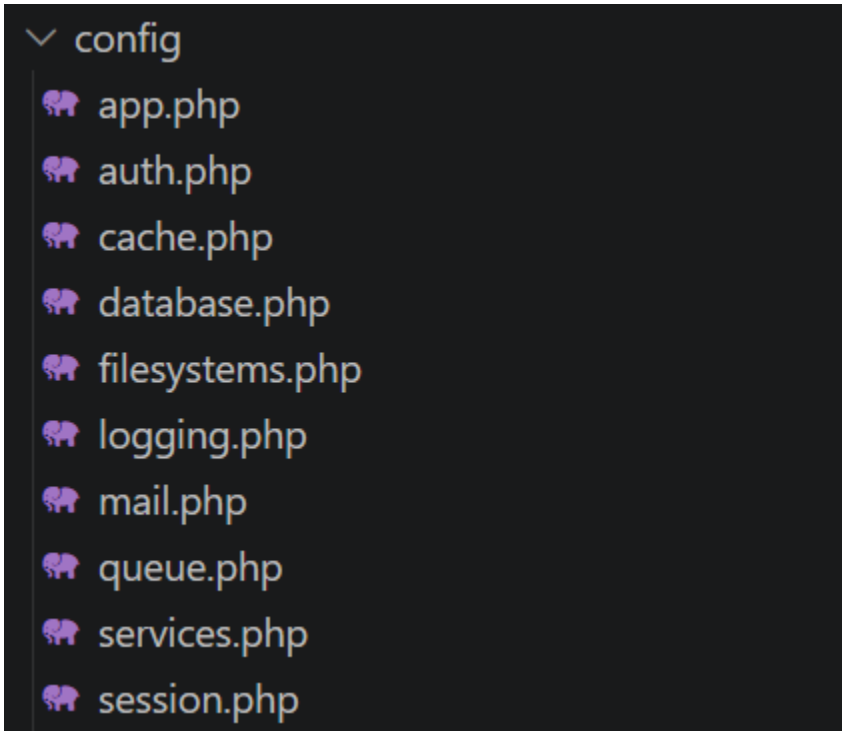
c. app/Models/users.php ← Model Eloquent (representasi tabel)



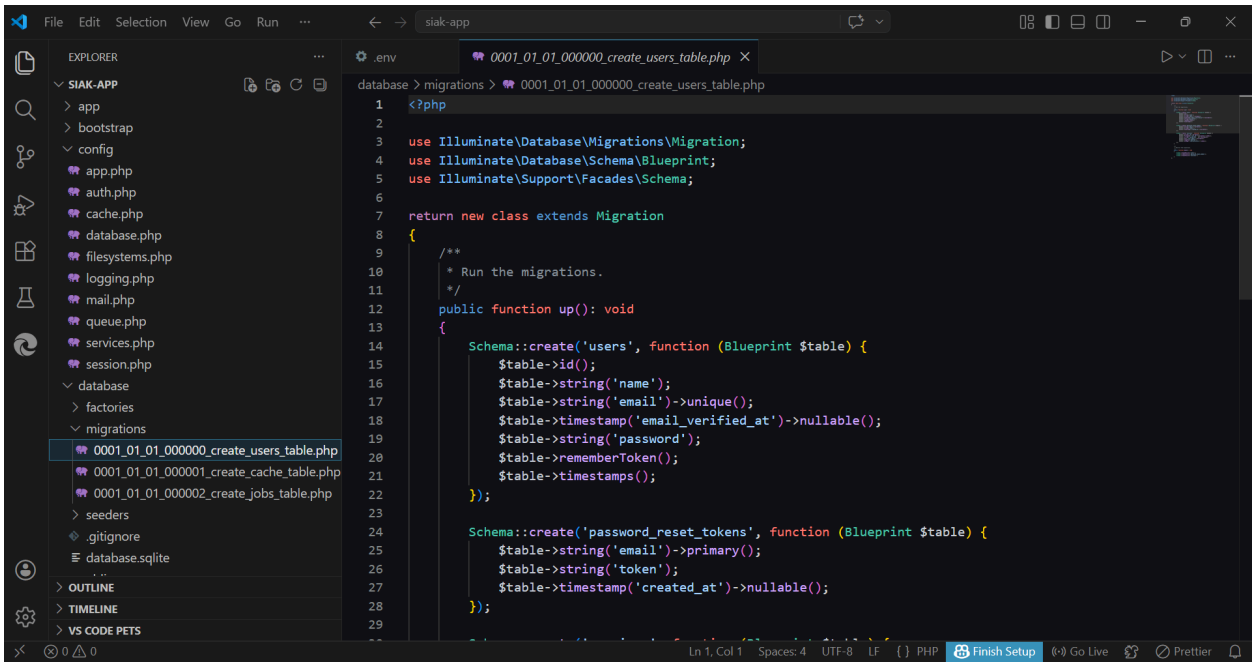
d. bootstrap/ ← Bootstrap aplikasi & cache



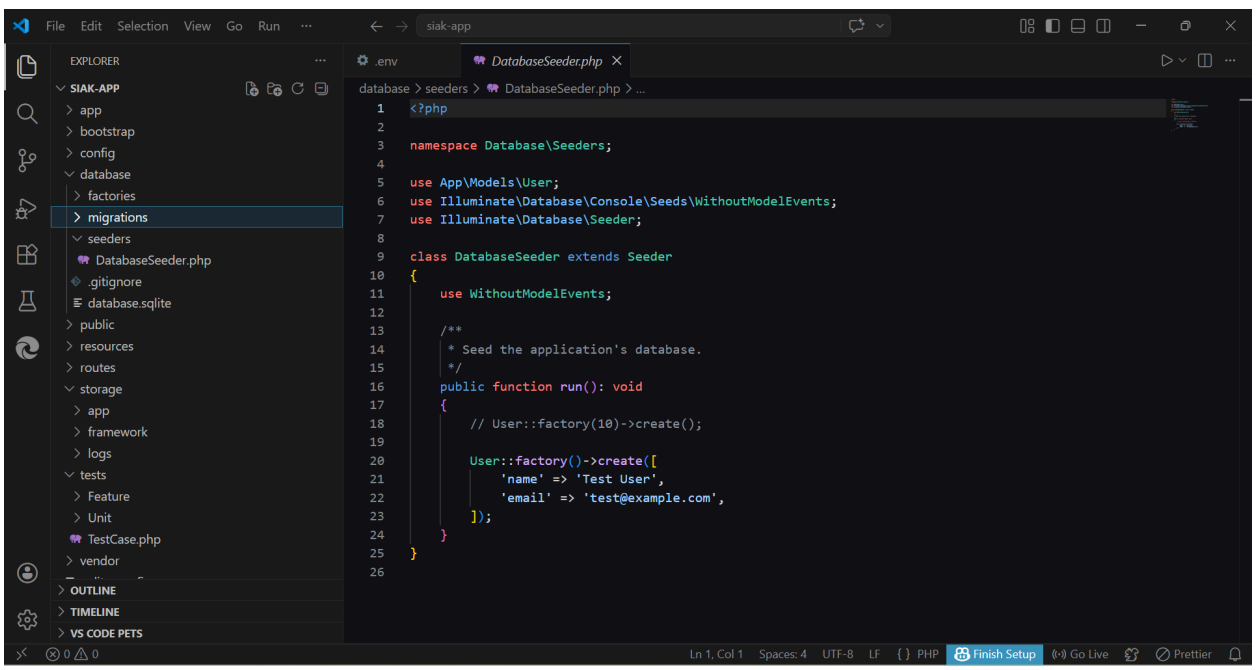
e. config/ ← File konfigurasi (database, mail, dll)



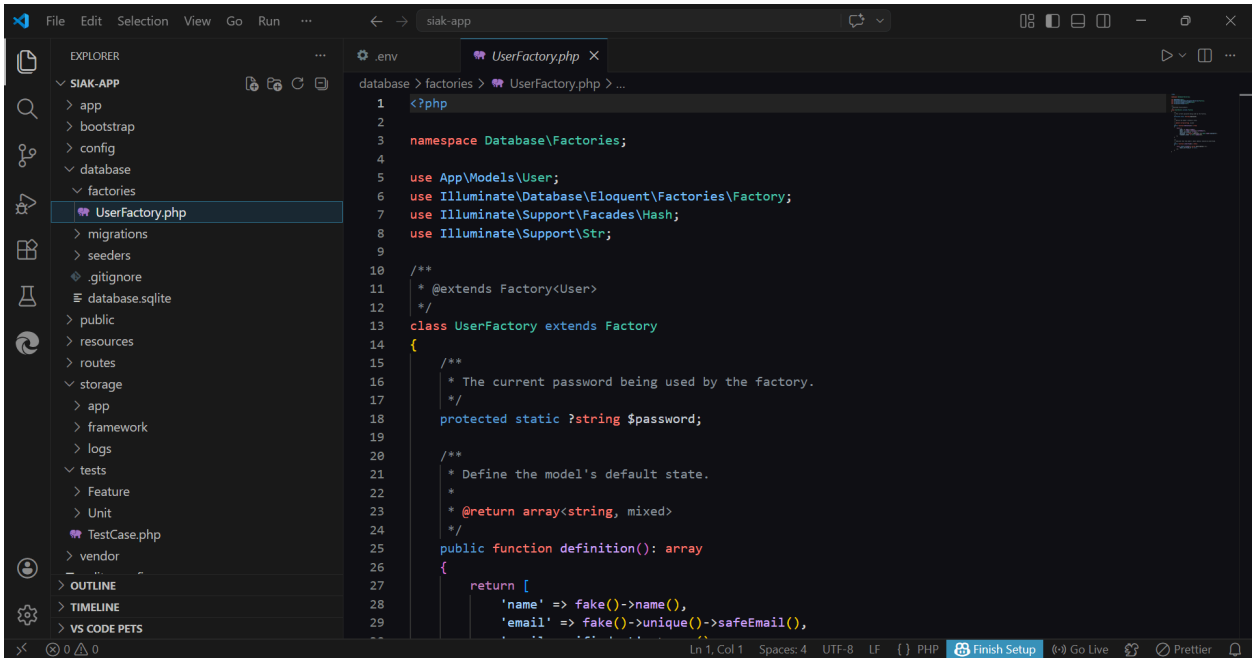
f. database/migrations ← Definisi skema tabel



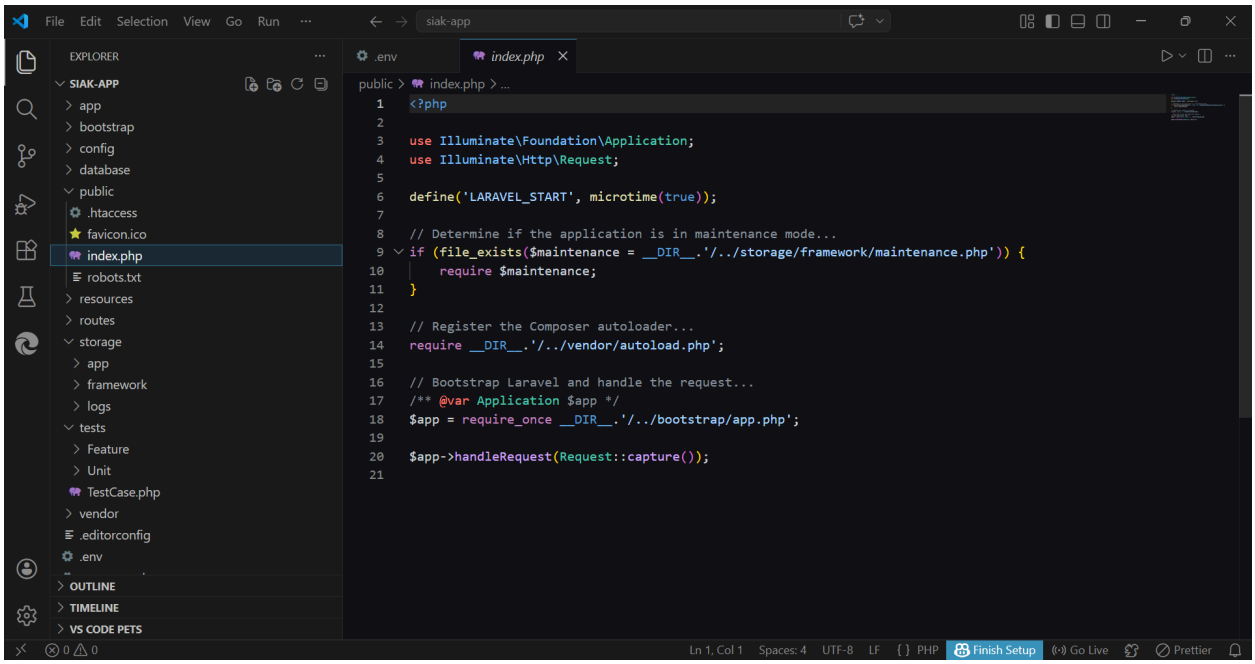
g. database/seeders ← Data awal / dummy



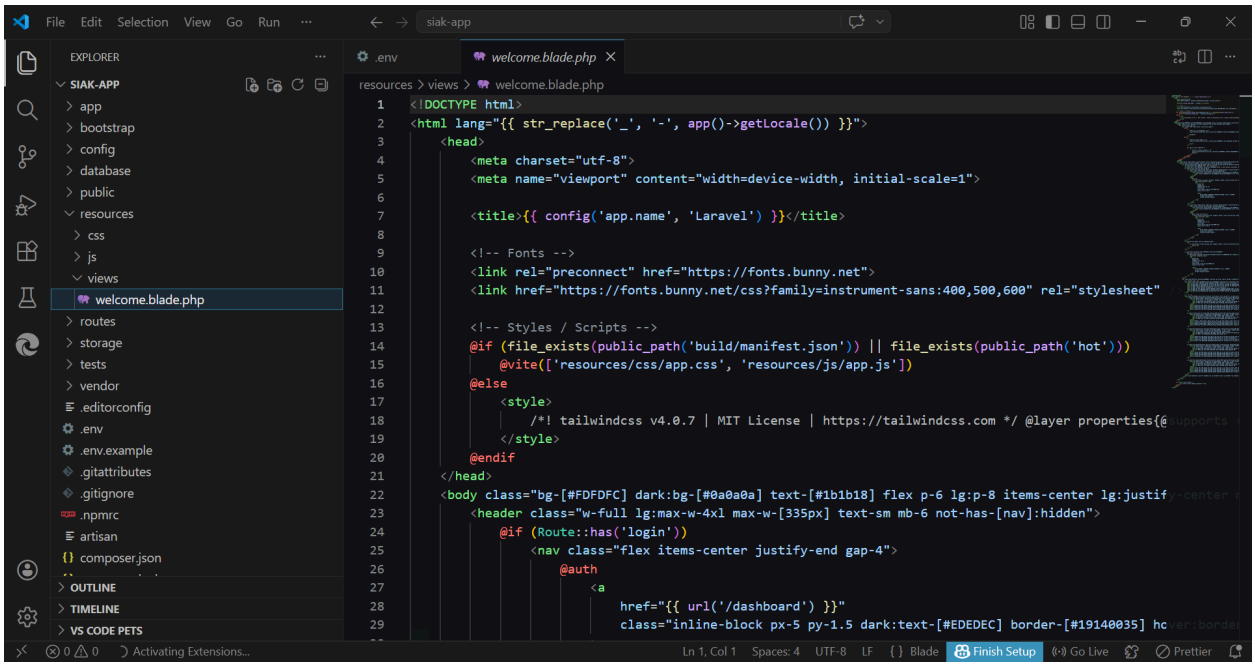
h. database/factories ← Factory untuk testing



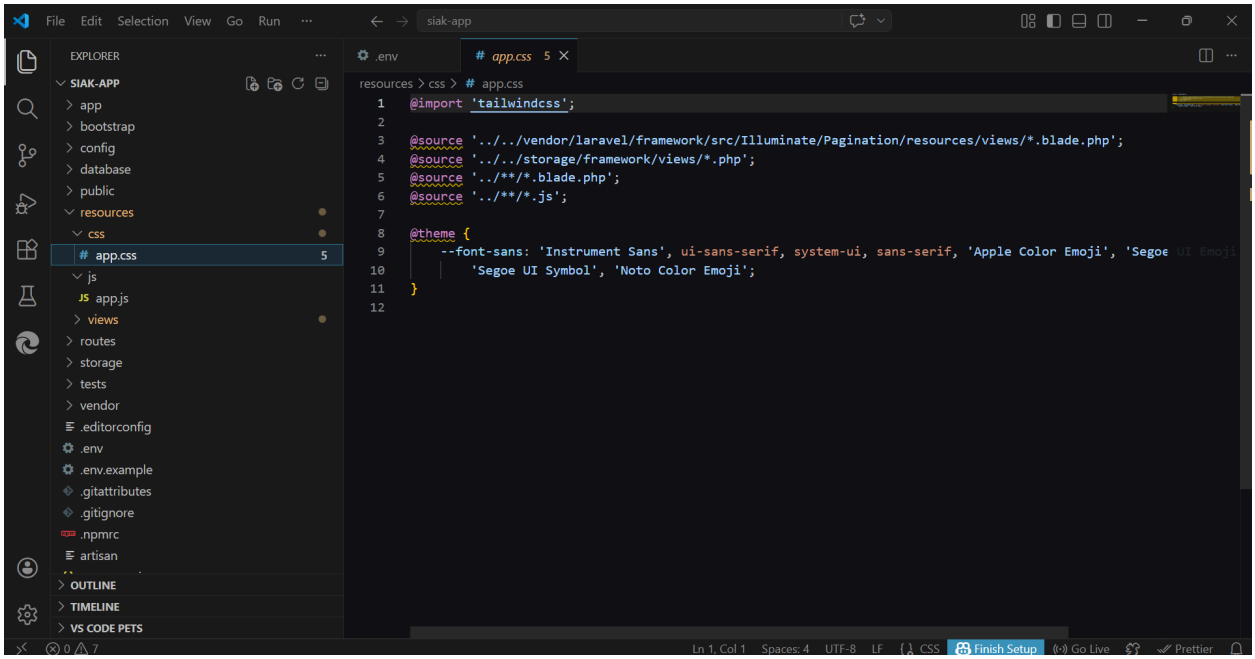
i. public/index.php ← Document root web server



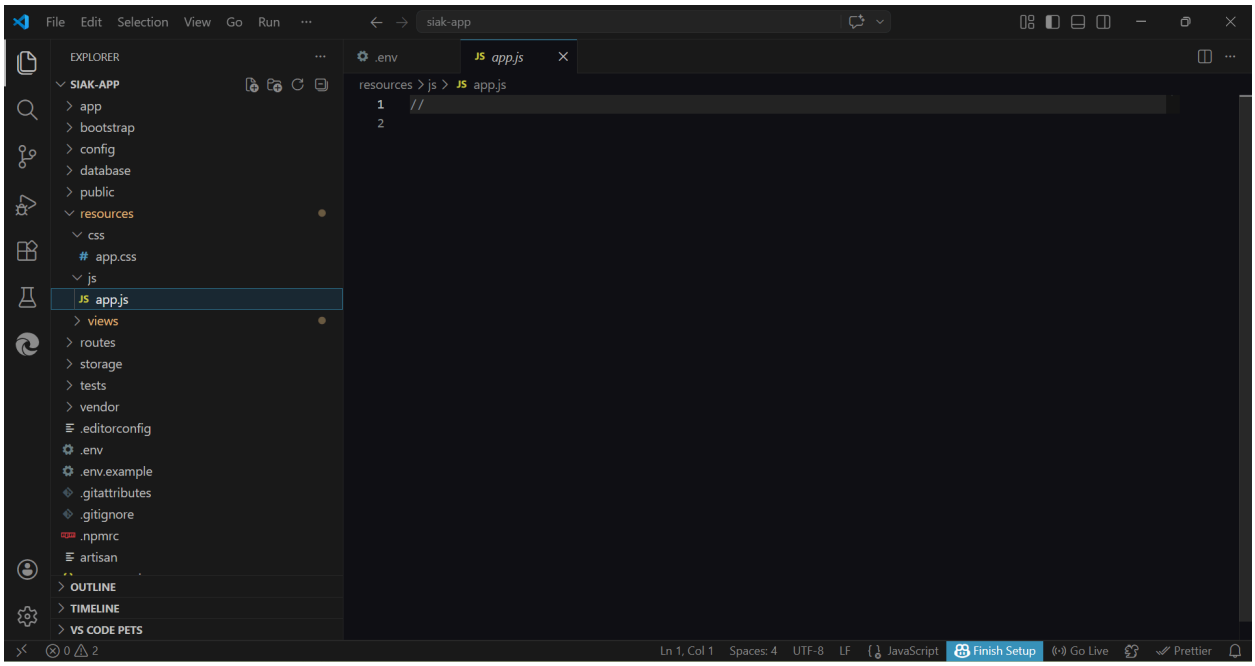
j. resources/views ← Blade template (file .blade.php)



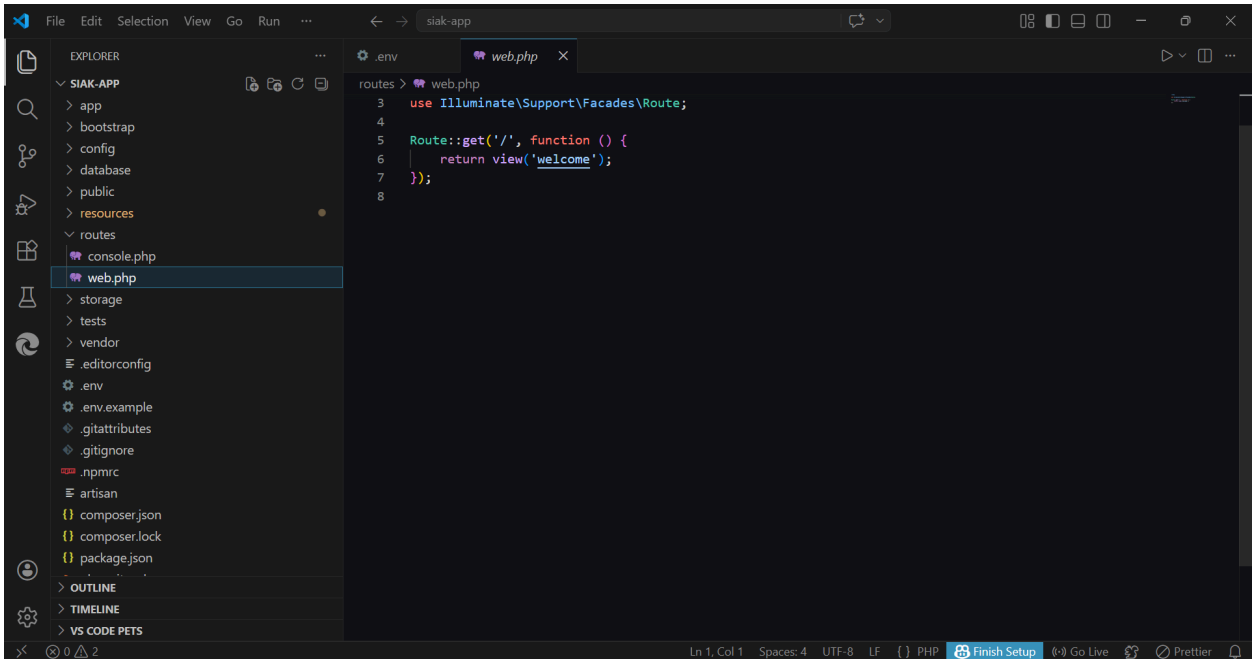
k. resources/css ← CSS sumber



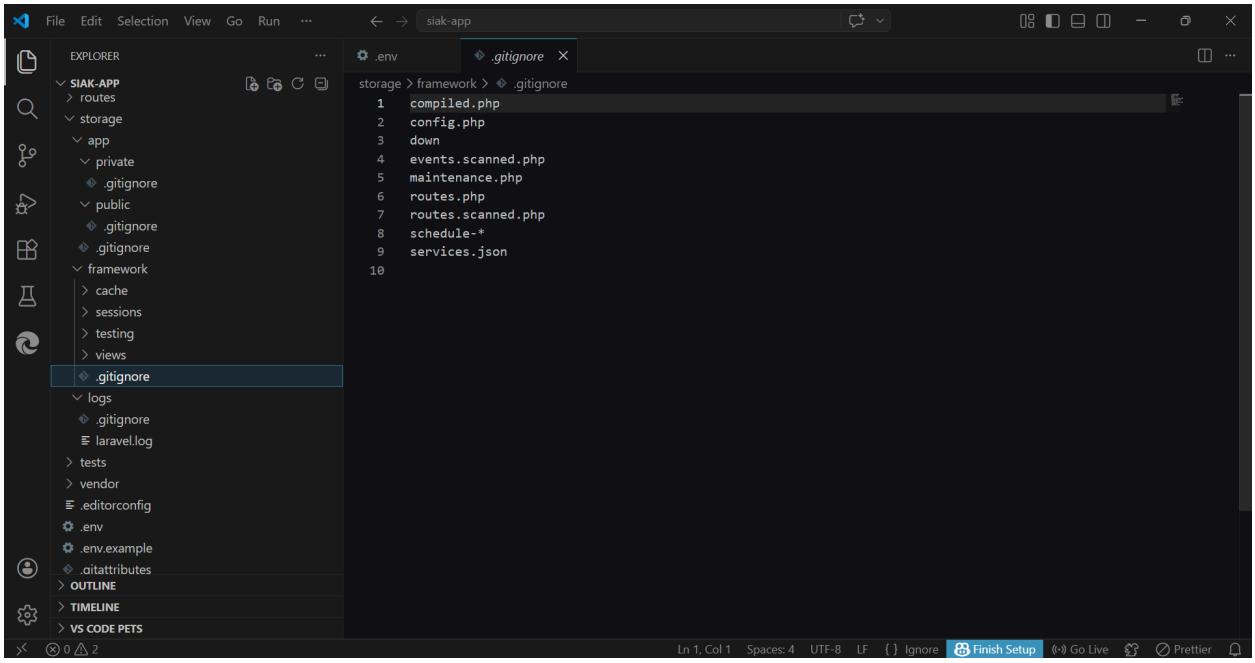
l. resources/js ← JavaScript sumber



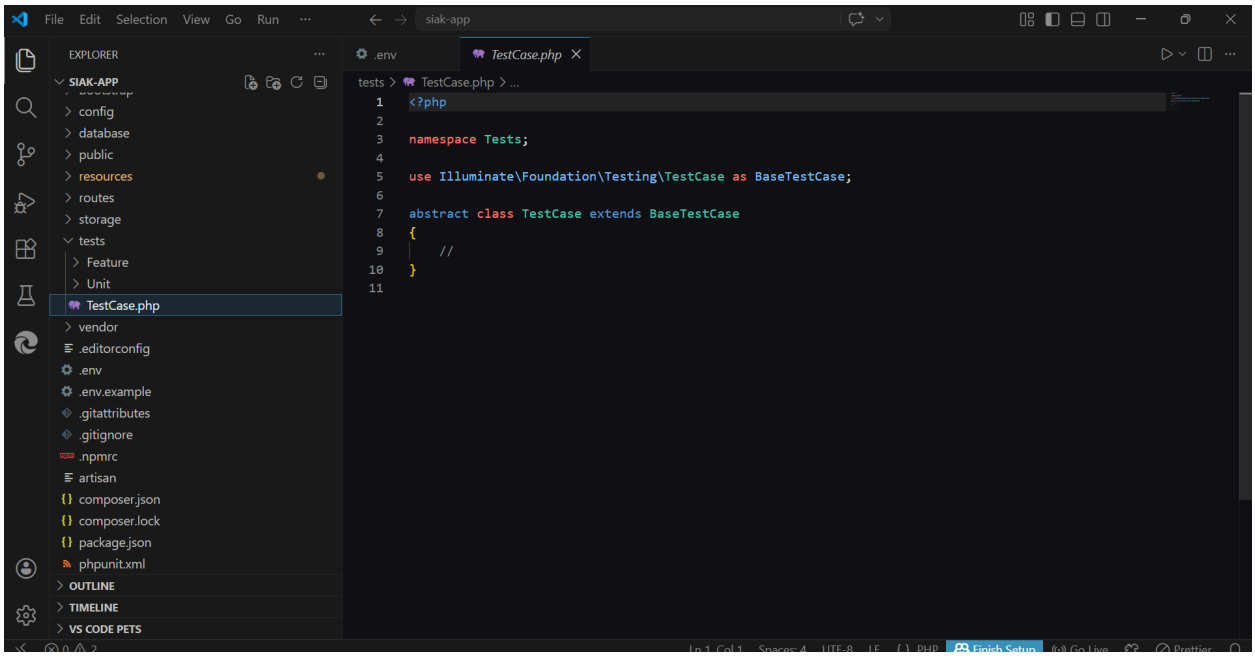
m. routes/web.php ← Routing untuk halaman web



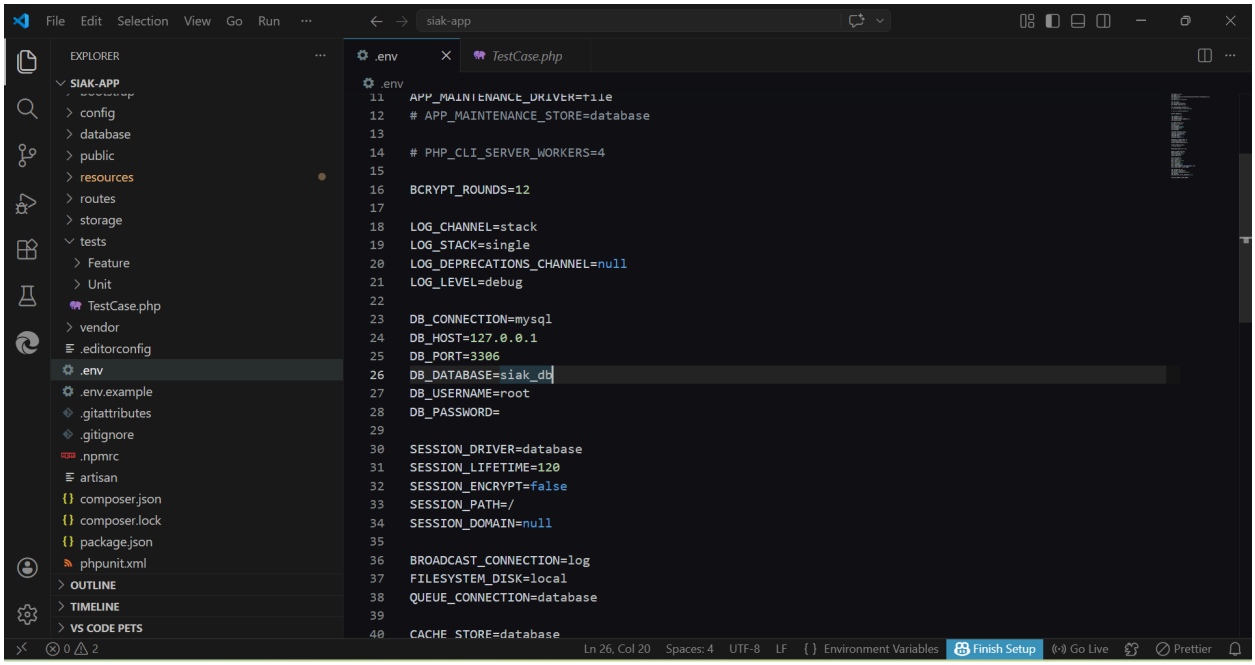
n. storage/ ← File upload, log, cache



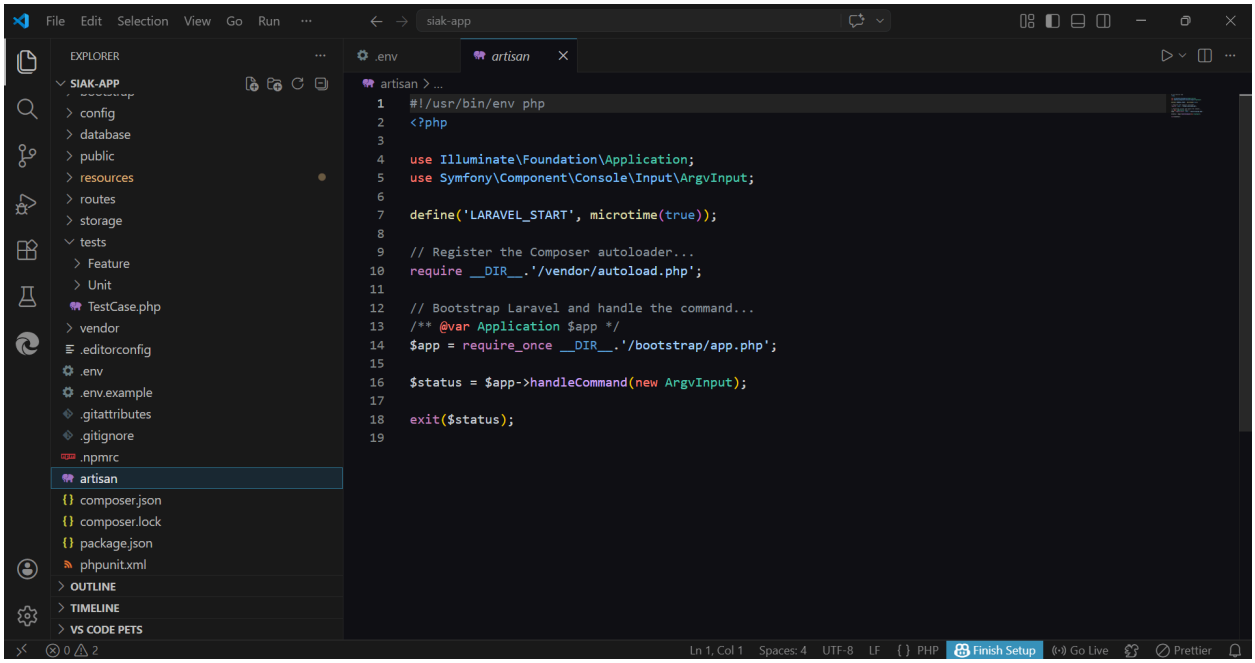
o. tests/ ← Unit & feature tests



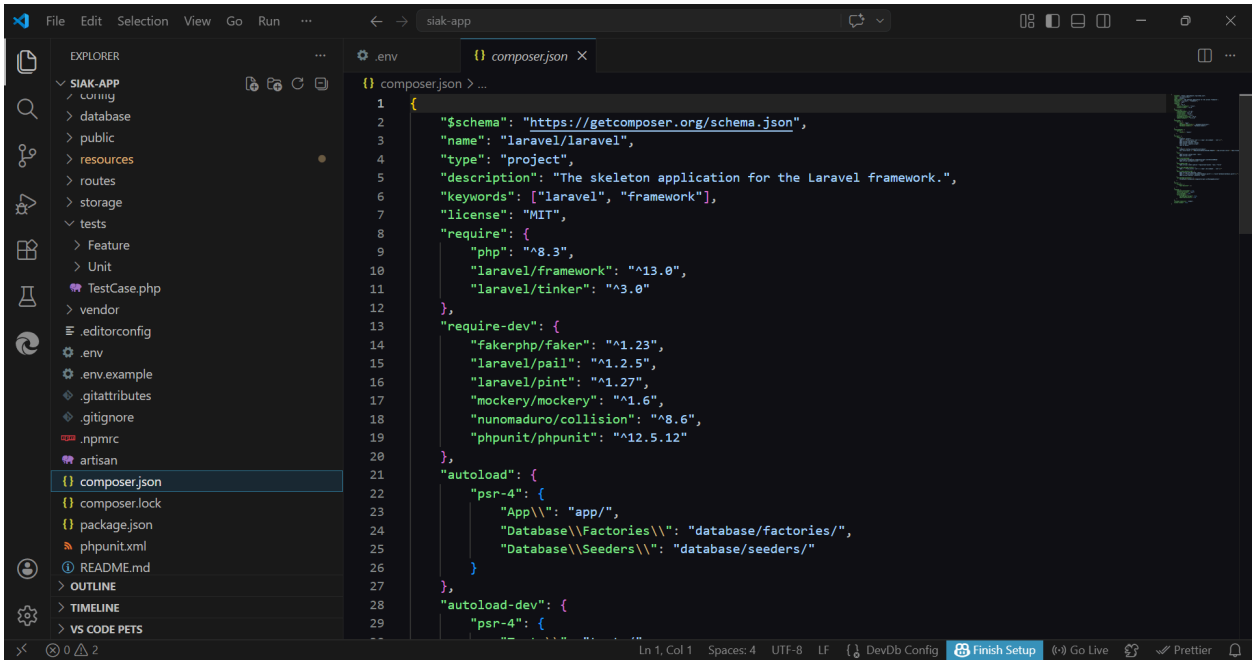
p. .env ← Konfigurasi environment



q. artisan ← Laravel CLI

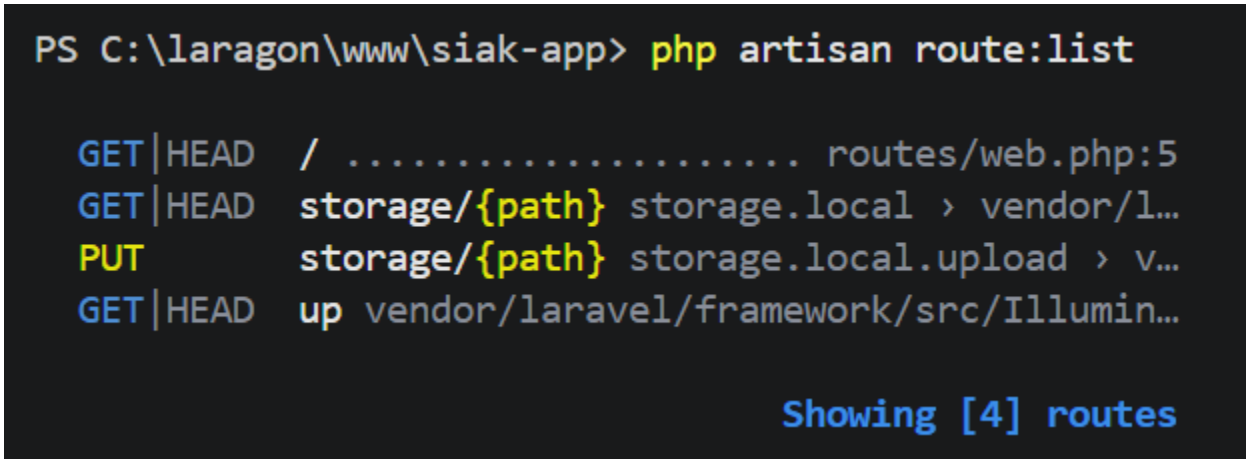


r. composer.json ← Dependensi PHP



9. Jalankan php artisan route:list dan catat route apa saja yang sudah terdaftar

Perintah `php artisan route:list` digunakan untuk melihat daftar route yang tersedia pada aplikasi Laravel.



Tugas 2

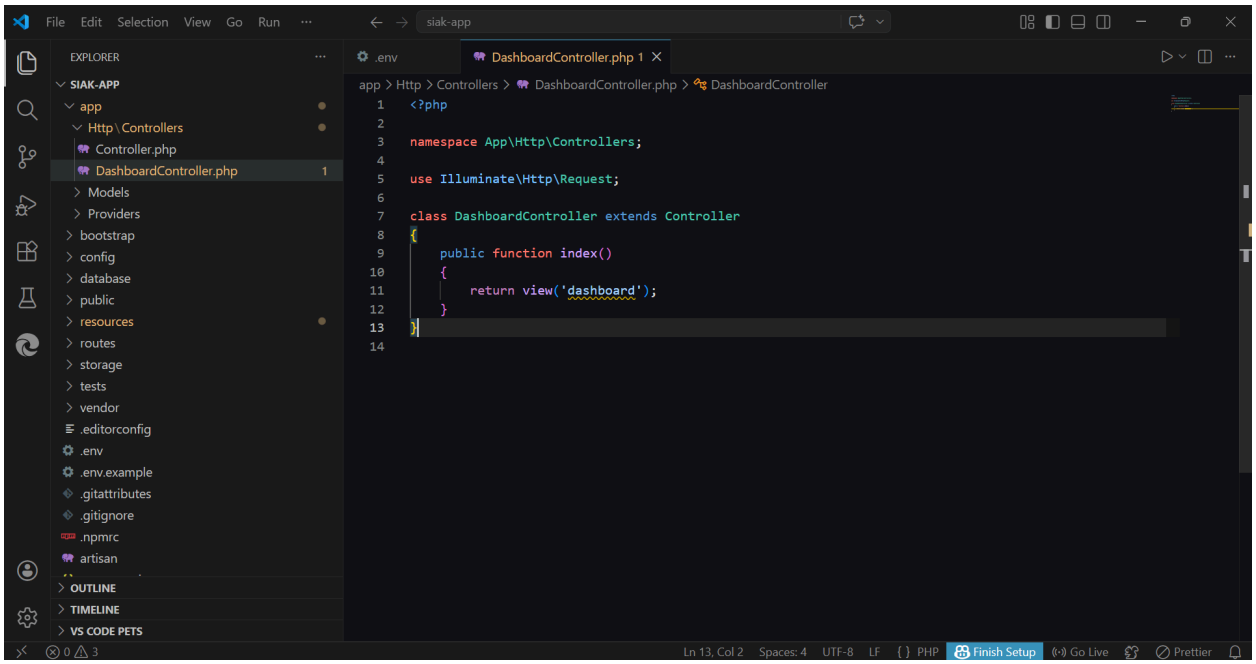
1. Buat Controller DashboardController: php artisan make:controller DashboardController

```
PS C:\laragon\www\siak-app> php artisan make:controller DashboardController

  INFO  Controller [C:\laragon\www\siak-app\app\Http\Controllers\DashboardController.php] created successfully.
```

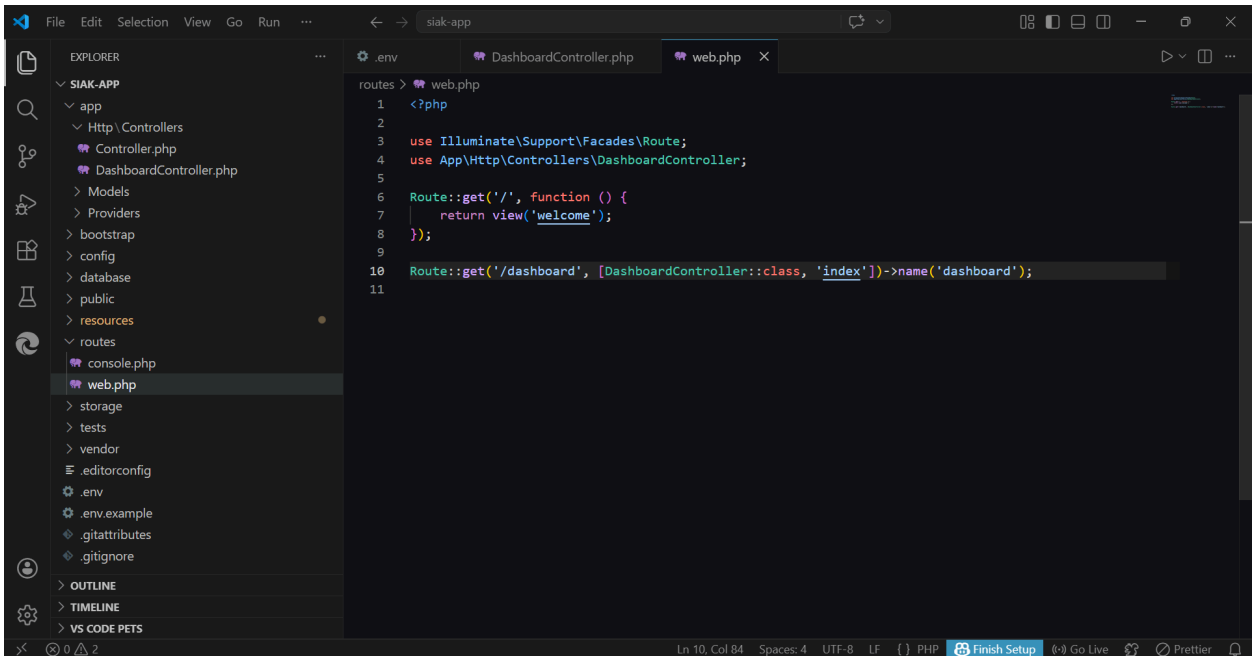
Controller dibuat dengan nama DashboardController yang nantinya digunakan untuk mengatur logika halaman dashboard.

2. Tambahkan method index() yang mengembalikan view('dashboard') di DashboardController



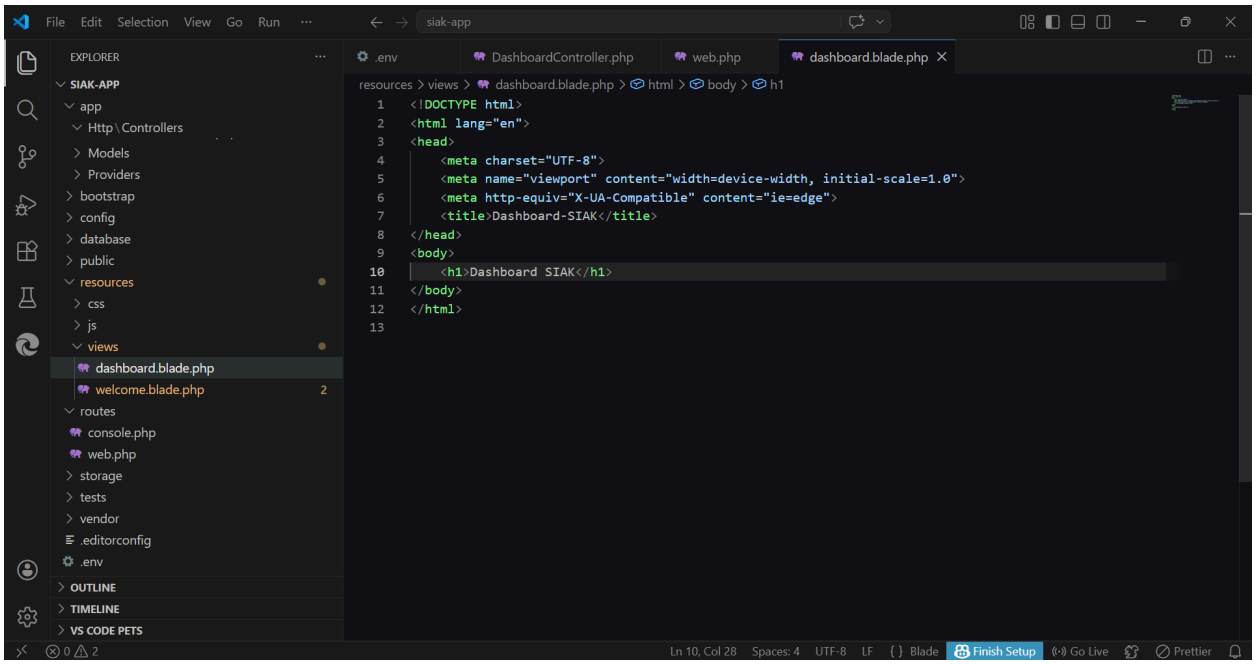
Method `index()` ditambahkan pada controller untuk mengembalikan tampilan halaman dashboard.

3. Daftarkan route di routes/web.php: Route::get('/dashboard', [DashboardController::class, 'index'])->name('dashboard')



Route ditambahkan untuk menghubungkan URL `/dashboard` dengan controller DashboardController agar halaman bisa diakses melalui browser.

4. Buat file view sementara resources/views/dashboard.blade.php berisi '<h1>Dashboard SIAK</h1>'



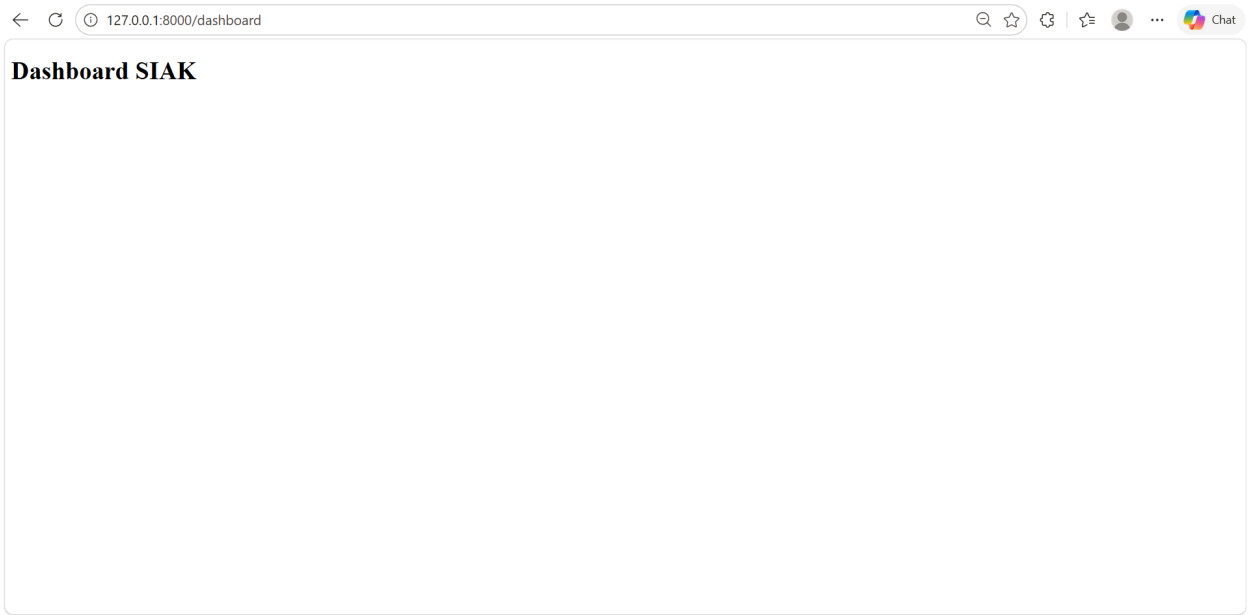
File view dibuat untuk menampilkan halaman dashboard dengan isi sederhana sebagai tampilan awal.

5. Akses http://127.0.0.1:8000/dashboard di browser pastikan tampil

```
PS C:\laragon\www\siak-app> php artisan serve

  INFO  Server running on [http://127.0.0.1:8000].

  Press Ctrl+C to stop the server
```



Halaman dashboard diakses melalui browser untuk memastikan route dan controller sudah berjalan dengan baik.

6. Buat route /tentang dengan Route::view('/tentang', 'tentang') dan buat view-nya

File Edit Selection View Go Run

siak-app

EXPLORER

SIAK-APP

- config
- database
- public
- resources
  - css
  - js
  - views
    - dashboard.blade.php
    - tentang.blade.php
    - welcome.blade.php 2
- routes
  - console.php
  - web.php
- storage
- tests
- vendor
- .editorconfig
- .env
- .env.example
- .gitattributes
- .gitignore
- .npmrc

OUTLINE

TIMELINE

VS CODE PETS

.env | DashboardController.php | web.php | tentang.blade.php | welcome.blade.php 2

routes > web.php

```php
<?php

use Illuminate\Support\Facades\Route;
use App\Http\Controllers\DashboardController;

Route::get('/', function () {
    return view('welcome');
});

Route::get('/dashboard', [DashboardController::class, 'index'])->name('dashboard');

Route::view('/tentang', 'tentang');
```

0 2 | Ln 12, Col 36 | Spaces: 4 | UTF-8 | LF | { } PHP | Finish Setup | Go Live | Prettier

File Edit Selection View Go Run

siak-app

EXPLORER

SIAK-APP

- config
- database
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- storage
- tests
- vendor
- .editorconfig
- .env
- .env.example
- .gitattributes
- .gitignore
- .npmrc

OUTLINE

TIMELINE

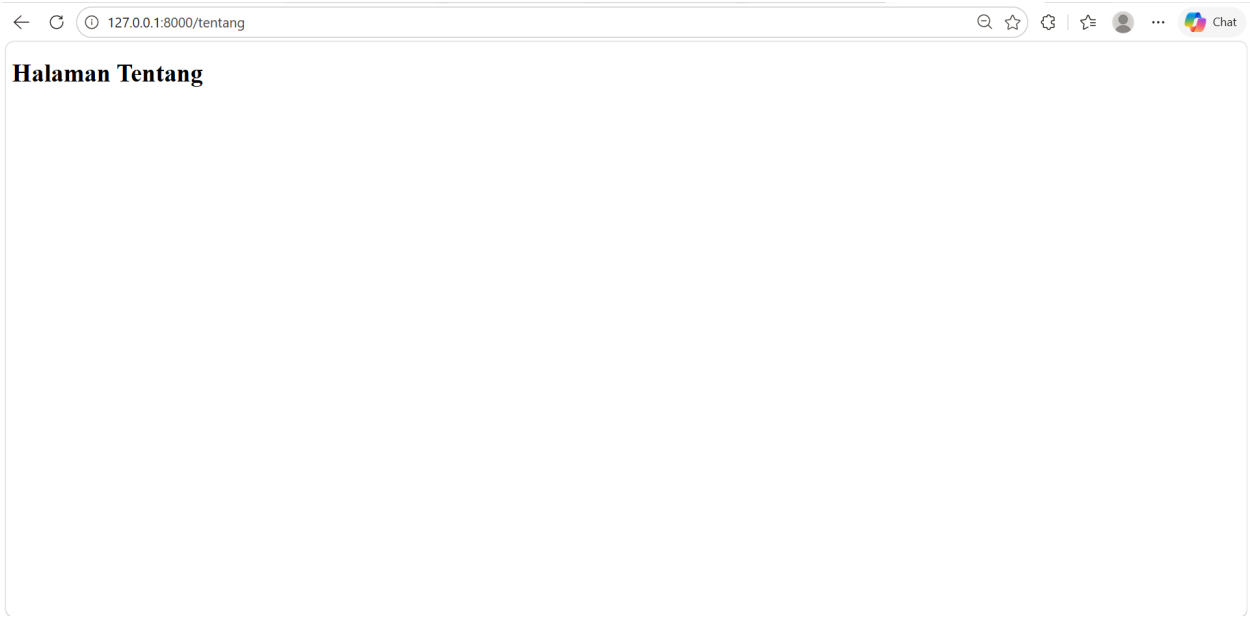
VS CODE PETS

.env | DashboardController.php | web.php | tentang.blade.php | welcome.blade.php 2 | dashboa

resources > views > tentang.blade.php > html > body > h1

```html
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <meta http-equiv="X-UA-Compatible" content="ie=edge">
    <title>Tentang-SIAK</title>
</head>
<body>
    <h1>Halaman Tentang</h1>
</body>
</html>
```

0 2 | Ln 10, Col 29 | Spaces: 4 | UTF-8 | LF | { } Blade | Finish Setup | Go Live | Prettier



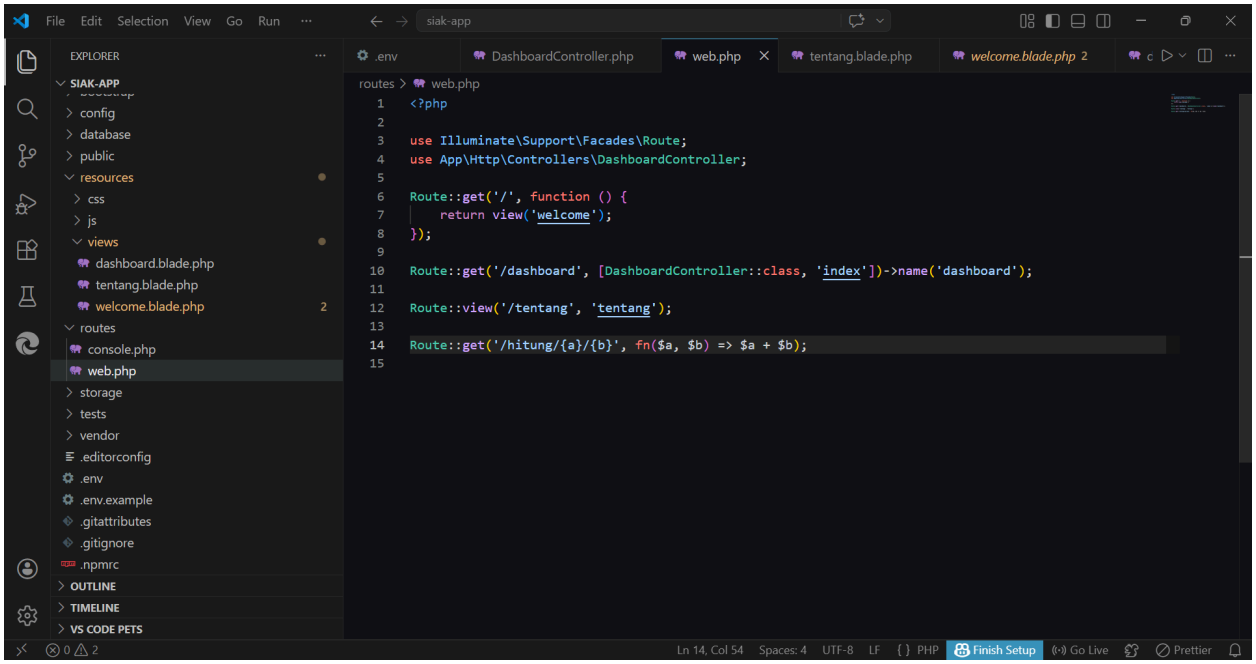
Route `/tentang` dibuat menggunakan Route::view untuk langsung menampilkan halaman tanpa melalui controller.

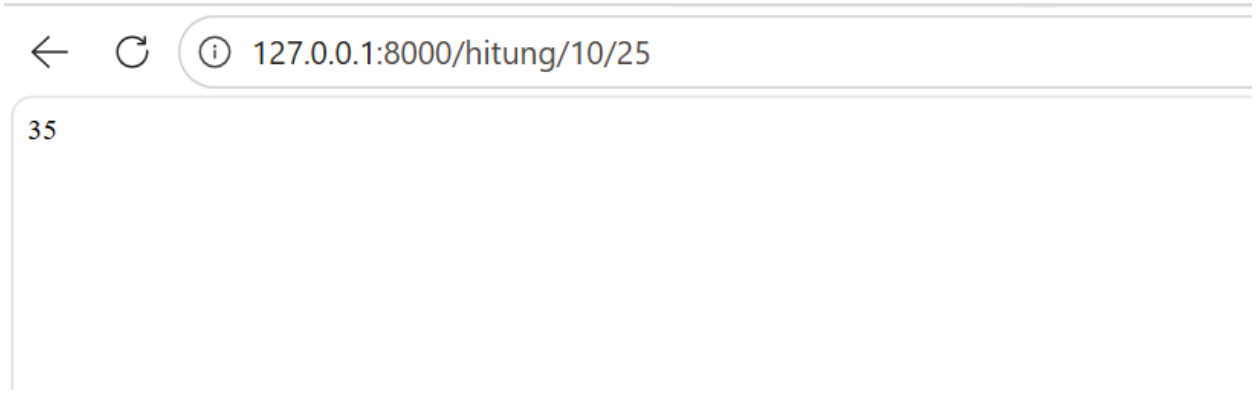
7. Jalankan php artisan route:list - amati semua route yang terdaftar

```
PS C:\laragon\www\siak-app> php artisan route:list

  GET|HEAD  / ..................................................................... routes/web.php:6
  GET|HEAD  dashboard ........................................................ dashboard › DashboardController@index
  GET|HEAD  storage/{path} storage.local › vendor/laravel/framework/src/Illuminate/Filesystem/FilesystemServicePr…
  PUT       storage/{path} storage.local.upload › vendor/laravel/framework/src/Illuminate/Filesystem/FilesystemSe…
  GET|HEAD  tentang ...............................................................................................
  GET|HEAD  up ....... vendor/laravel/framework/src/Illuminate/Foundation/Configuration/ApplicationBuilder.php:223

                                                                                          Showing [6] routes
```

Route list dijalankan kembali untuk melihat semua route yang telah dibuat, termasuk dashboard dan tentang.

8. Coba buat Route::get('/hitung/{a}/{b}', fn($a, $b) => $a + $b) dan akses /hitung/10/25





Route dinamis dibuat dengan parameter untuk melakukan operasi penjumlahan, sehingga dapat menghitung langsung dari URL.